

Computational Wind Engineering

Buffeting analysis

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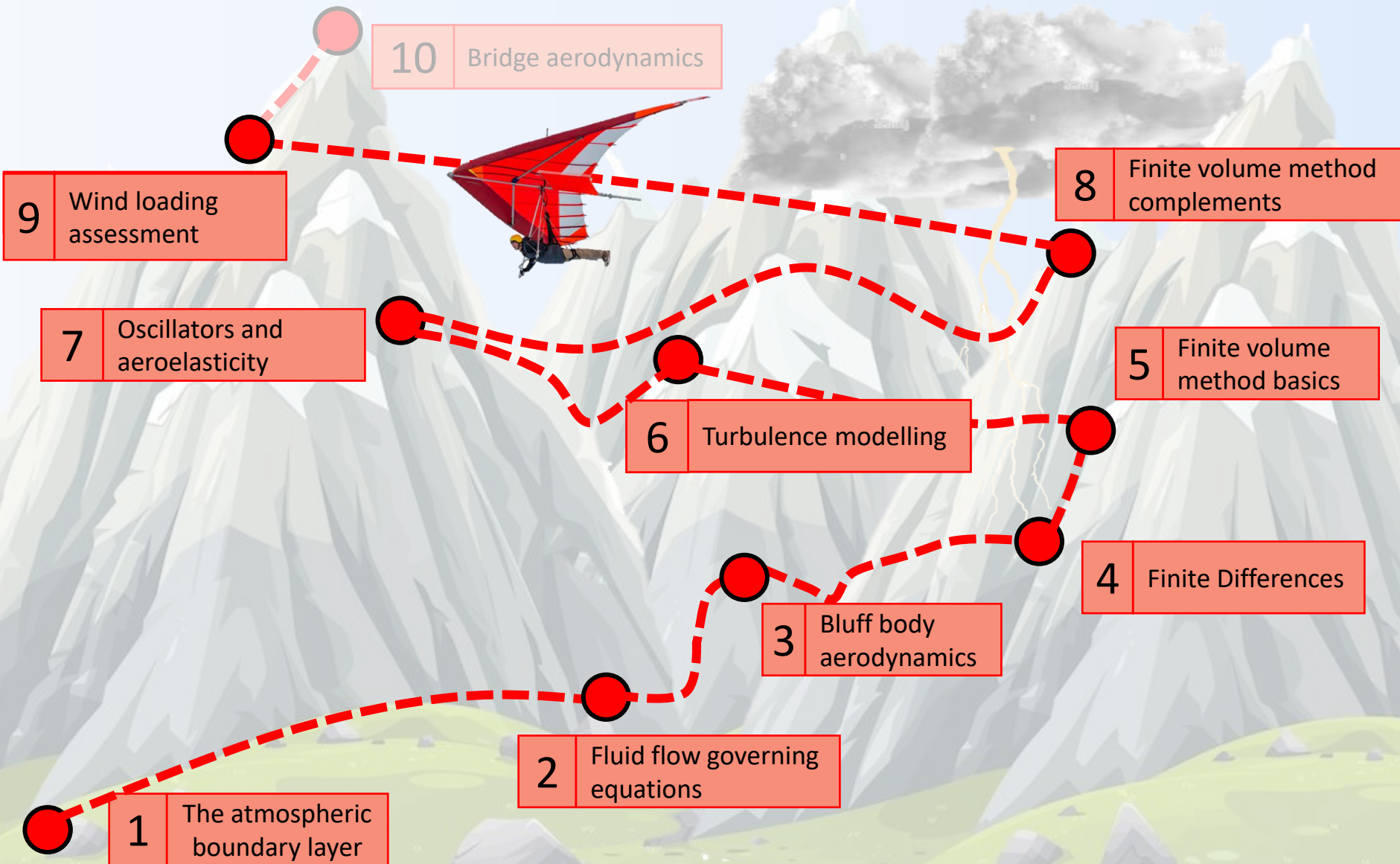
- Buffeting analysis
- Point-like structures: along-wind
- Peak factor
- Point-like structures: cross-wind
- Line-like structures
- Block-maxima approach

Map of the course

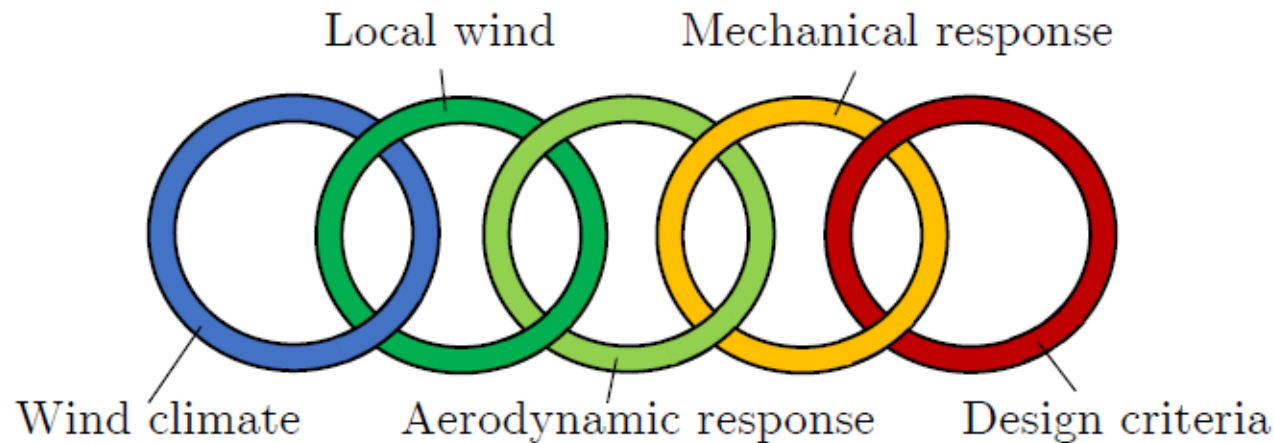
Wind Engineering

Fluid Dynamics

Numerical methods



We are now ready to consider again Davenport's wind load chain



The only step performed using Wind tunnel tests or CFD based simulations are the 2nd and the 3rd.

We now consider how we can pass from the aerodynamic response to the mechanical response, and the fulfilment of design criteria.

In general, **once pressures are known, it is possible to apply them to a mechanical model of the structure.**



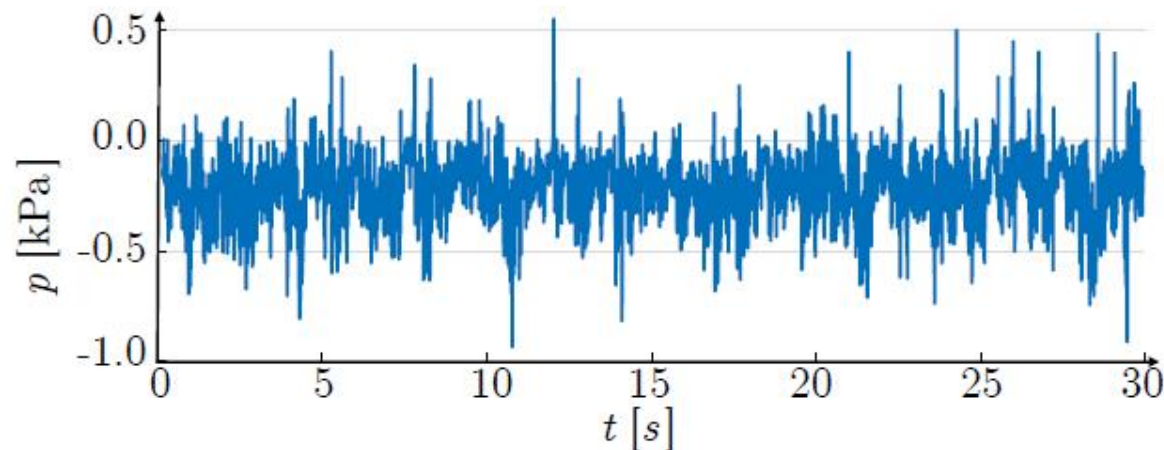
(a)



Pressure taps

Piping system for
pressure
measurement

(b)



However, also this simple passage requires appropriate considerations. In particular, we measure pressure in scaled models and convert it to pressure coefficients. Then, we revert the process

$$p(t) = qc_p(t) \quad \text{where} \quad q = \frac{1}{2}\rho U_{rf}^2.$$

Pay attention, also time need to be rescaled!

In particular, let us assume that the recorded signals are of length T_m , the signal time duration in model scale. The non-dimensional time unit is

$$S_m = B_m/U_{rm}.$$

where B_m is the model length scale and U_{rm} is the model reference velocity. The total number of non dimensional time units will be

$$T_m/S_m$$

The number of non-dimensional time units will be the same in full and reduced scale, so we have

$$\frac{T_m}{S_m} = \frac{T_f}{S_f} \quad \Rightarrow \quad \frac{T_m U_m}{B_m} = \frac{T_f U_f}{B_f} \quad \Rightarrow \quad T_f = T_m \frac{B_f U_m}{B_m U_f},$$

which transforms the model signal time length into full-scale time length. More in general

$$t_f = t_m \frac{B_f}{B_m} \frac{U_m}{U_f},$$

which establishes a relation between model time and full-scale time. Notice that B_f/B_m is the geometrical scale, e.g 100 or 400.

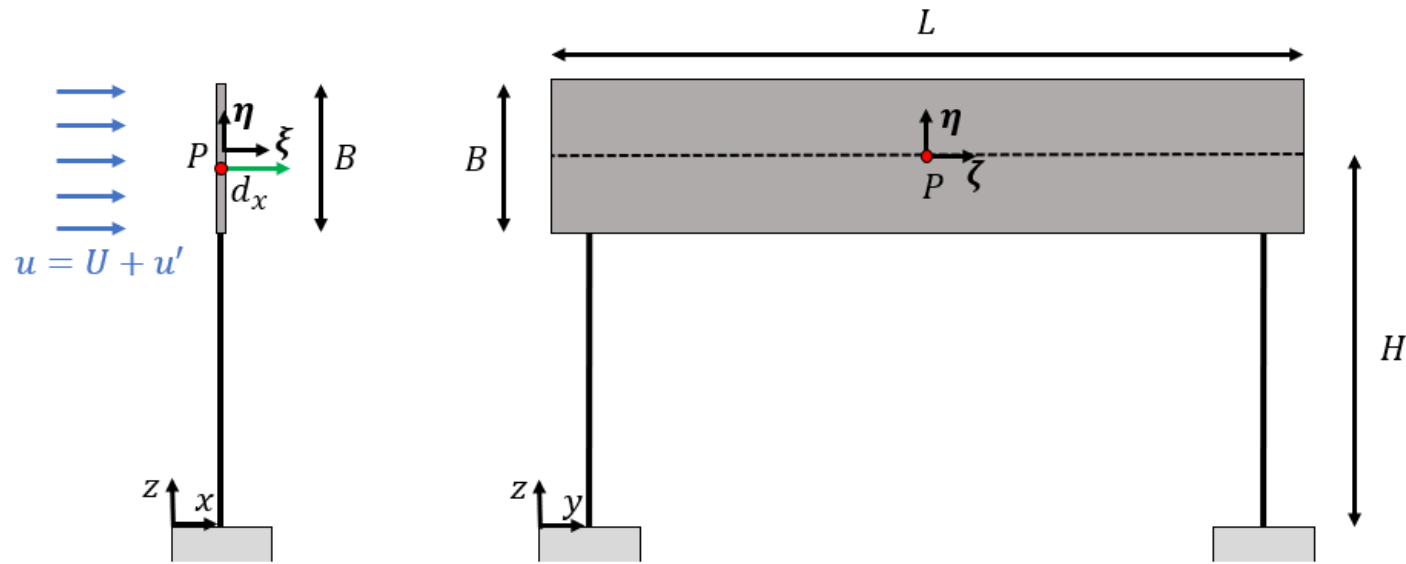
Once pressure is calculated from c_p and time is transformed, the resulting time series can be applied to the structural model.

Elements which must be always checked are

- The length of the time histories. The standard sample is 10 min full scale, to be repeated multiple times. They must be long enough to extract meaningful statistics.
- Keep blockage under 5% in WT or simulations;
- Also the velocity profile and incoming turbulence must be scaled properly;
- Consider wind coming from all directions, usually every 10-30 degrees;
- The structure is usually rigid, so disregarding aeroelastic effects.

Assume you can calculate the structural response and obtain a time history of all design quantities. How do we select a proper design value? We need to rely on statistics and Extreme Value Theory.

To shed some more light on buffeting analysis, we now consider a simple case: a sign board.



In general, if there is no incoming turbulence we have

$$F_{\xi} = \frac{1}{2} \rho U^2 L B C_{\xi},$$

We now consider that **the velocity in P is representative of the velocity everywhere in the sign proximity** (the structure is small compared to the incoming turbulence).

Under this assumption, the force can be calculated relying on the velocity measured in P, and the corresponding displacement will be

$$d_x(t) = \frac{1}{k_s} \frac{1}{2} \rho u_P(t)^2 BLC_\xi,$$

being k_s the structural stiffness. From now on we disregard the subscript P. Developing the square

$$d_x(t) = \frac{1}{k_s} \frac{1}{2} \rho (U + u'(t))^2 BLC_\xi = \frac{1}{2k_s} \rho [U^2 + 2Uu'(t) + u'(t)^2] BLC_\xi,$$

$$d_x(t) = \frac{1}{2k_s} \rho U^2 \left[1 + 2\frac{u'(t)}{U} + \left(\frac{u'(t)}{U} \right)^2 \right] BLC_\xi,$$

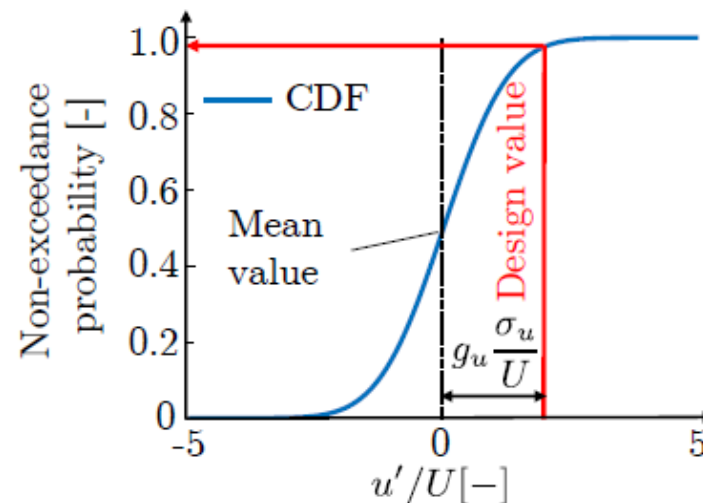
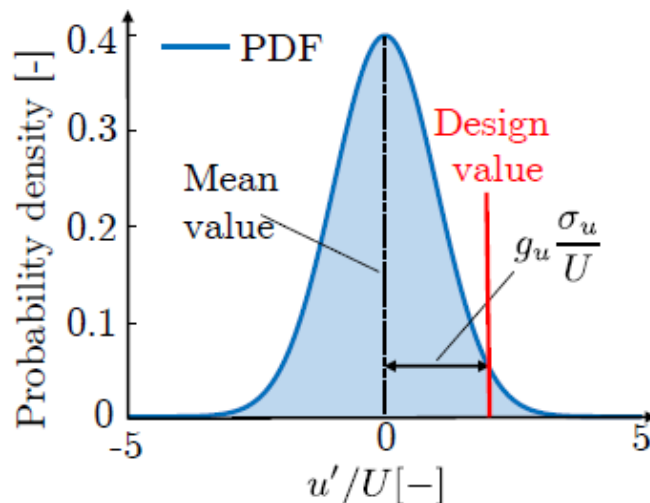
$$d_x(t) \approx \frac{1}{2k_s} \rho U^2 \left[1 + 2\frac{u'(t)}{U} \right] BLC_\xi.$$

which assumes small I_u

This formula suggests two approaches for the evaluation of the design values

$$d_x(t) \approx \frac{1}{2k_s} \rho U^2 \left[1 + 2 \frac{u'(t)}{U} \right] BLC_\xi.$$

- 1) Produce $u'(t)$ compatible with the one on site, calculate the response and extract a conveniently high value (to be discussed how);
- 2) Select a conveniently high (unlikely) value of $u'(t)$, apply the formula and directly obtain an extreme design value.



Adopting the second approach

$$\hat{u}' = g_u \sigma_u = g_u U I_u,$$

where the hat denotes a **maximum peak value**

$$\frac{\hat{u}'}{U} = g_u \frac{\sigma_u}{U} = g_u I_u,$$

so that

$$\hat{d}_x = \frac{1}{2k_s} \rho U^2 \left(1 + 2 \frac{\hat{u}'}{U} \right) BLC_\xi = \frac{1}{2k_s} \rho U^2 (1 + 2g_u I_u) BLC_\xi,$$

or also

$$\hat{d}_x = D_x (1 + 2g_u I_u),$$

where D_x is the along wind response to the time averaged wind, U .

We can take a different perspective on the previously developed results

$$d_x(t) \approx \frac{1}{2k_s} \rho U^2 \left[1 + 2 \frac{u'(t)}{U} \right] BLC_\xi = D_x + \frac{1}{2k_s} \rho U^2 2 \frac{u'(t)}{U} BLC_\xi,$$

We might in fact notice that

$$\sigma_{d_x} = \frac{1}{2k_s} \rho U^2 2 \frac{\sigma_u}{U} BLC_\xi,$$

So that the previously presented approach based on the peak factor now reads

$$\hat{d}_x = \frac{1}{2k_s} \rho U^2 (1 + 2g_u I_u) BLC_\xi = D_x + g_u \sigma_{d_x},$$

where now the peak factor is applied to the standard deviation of the response.

In general, we thus have the following possibilities to extract a design value using peak factors

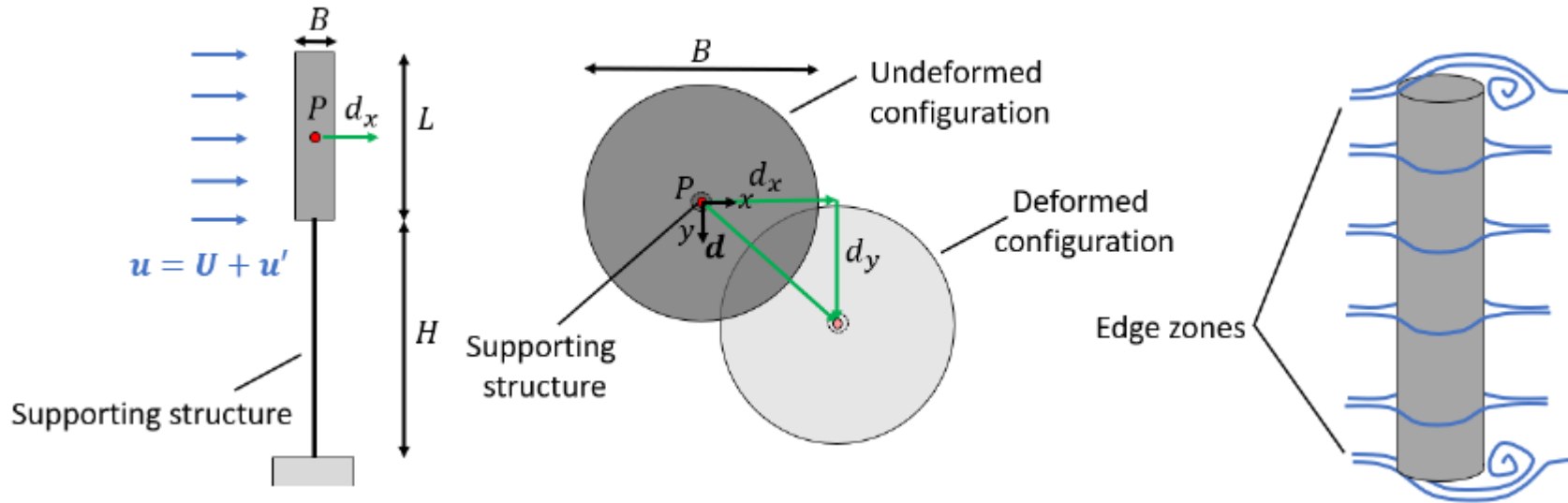
- A1 - the cause of the excitation, e.g. the velocity field using g_u in this case;
- A2 - the time-average structural response, using $1 + 2g_u I_u$;
- A3 - the structural response itself, using g_u in this case.

The second approach is known as Davenport's peak loading factor. Using a $g_u = 3.5$ leads to a non-exceedance probability of 99.976% (true only if Gaussain), and it is often adopted in codes of practice.

The term $2g_u$, taking $g_u = 3.5$, leads to 7, which is a number commonly appearing in codes and standards when calculating the, so called, peak dynamic pressure, i.e.

$$\hat{q} = \frac{1}{2} \rho \hat{u}^2 = \frac{1}{2} \rho (U + \hat{u}')^2 \approx \frac{1}{2} \rho U (1 + 2g_u I_u) = q(1 + 7I_u)$$

Let us now consider the following case, similar to an antenna



$$\mathbf{f}(t) = \frac{1}{2} \rho C_d B L |\mathbf{u}_P(t)| \mathbf{u}_P(t),$$

We disregard along wind fluctuations, which lead to the already seen along wind response

$$\mathbf{u}(t) = \begin{bmatrix} U \\ v'(t) \end{bmatrix}. \quad |\mathbf{u}(t)| = [U^2 + v'(t)^2]^{1/2} \approx U,$$

So that the forces acting on the antenna

$$\begin{bmatrix} f_x \\ f_y \end{bmatrix} \approx \frac{1}{2} \rho C_d B L U^2 \begin{bmatrix} 1 \\ \frac{v(t)}{U} \end{bmatrix}$$

so that the cross-wind response becomes

$$d_y(t) = \frac{1}{2k_s} \rho U^2 \frac{v(t)}{U} C_d B L.$$

It can be noticed that, in this case, the result is very similar to that previously obtained for the along wind response, but there is no mean part. **It is not possible to amplify the mean response!!**

Notice that in all previous examples we always assumed full-correlation of the velocity field!

Assume now the following situation, **in which the velocity field is not fully correlated**, i.e. the structure is not point-like. For each “stripe”

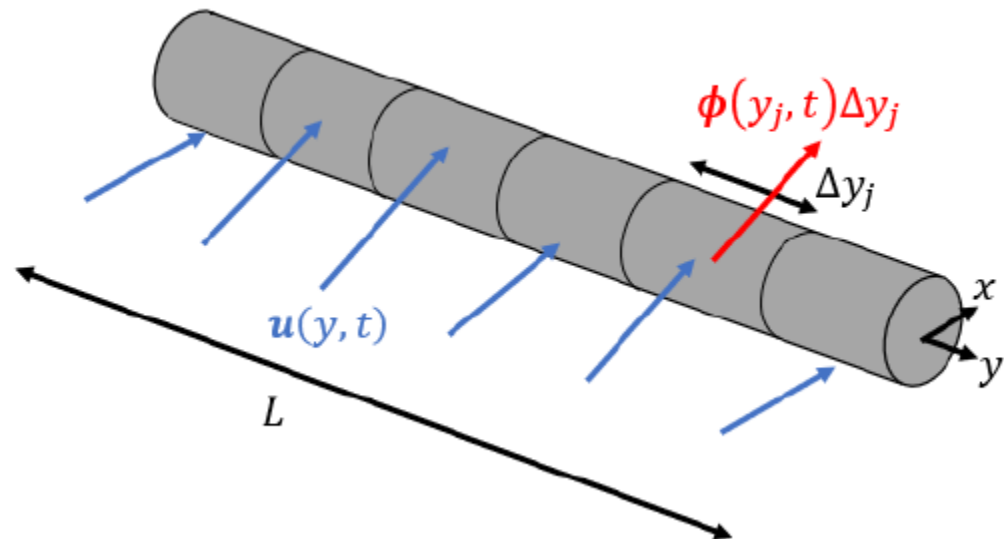
$$\phi(y, t) = \frac{1}{2} \rho C_d B |u(y, t)| u(y, t),$$

Now, the force acting on the whole structure is

$$f(t) = \sum_j^J \phi(y_j, t) \Delta y_j.$$

Let now consider a single component of $f(t)$, say $f_x(t)$.

We now wish to calculate, mean and standard deviation of the overall force, which are the necessary elements to evaluate the force peak value.



We start calculating the mean value of the overall force along the structure

$$\begin{aligned} F_x &= \frac{1}{T} \int_0^T f_x(t) dt = \frac{1}{T} \int_0^T \sum_{i=1}^J \phi_x(z_j, t) \Delta z_j dt, \\ &= \sum_{j=1}^J \frac{1}{T} \int_0^T \phi_x(z_j, t) dt \Delta z_j = \sum_{j=1}^J \Phi_x(z_j) \Delta z_j. \end{aligned}$$

which shows that the mean value of the overall force is calculated starting from mean value of the forces on the "stripes".

If we want to write it in a continuous fashion

$$F_x = \int_0^L \Phi_x(z) dz,$$

As regard the calculation of the standard deviation, we already observed that if a random variable is obtained by summing other random variables

$$\sigma_{f_x}^2 = \beta^T \underline{\Sigma}_{\phi_x} \beta,$$

being $\underline{\Sigma}_{\phi_x}$ the covariance matrix . In our case

$$\underline{\Sigma}_{\phi_x} = \begin{bmatrix} \sigma_{\phi_{x1}}^2 & \text{COV}(\phi_{x1}, \phi_{x2}) & \dots & \text{COV}(\phi_{x1}, \phi_{xJ}) \\ \text{COV}(\phi_{x1}, \phi_{x2}) & \sigma_{\phi_{x2}}^2 & \dots & \text{COV}(\phi_{x2}, \phi_{xJ}) \\ \dots & \dots & \dots & \dots \\ \text{COV}(\phi_{x1}, \phi_{xJ}) & \text{COV}(\phi_{x2}, \phi_{xJ}) & \dots & \sigma_{\phi_{xJ}}^2 \end{bmatrix}.$$

or in a continuous fashion

$$\sigma_{f_x}^2 = \int_0^L \int_0^L \text{COV}(\phi_x(\xi), \phi_x(\eta)) d\xi d\eta,$$

To obtain a better insight, let us consider the simple case

$$\sigma_{f_x}^2 = [\Delta z_1 \quad \Delta z_2] \begin{bmatrix} \sigma_{\phi_{x1}}^2 & \text{COV}(\phi_{x1}, \phi_{x2}) \\ \text{COV}(\phi_{x1}, \phi_{x2}) & \sigma_{\phi_{x2}}^2 \end{bmatrix} \begin{bmatrix} \Delta z_1 \\ \Delta z_2 \end{bmatrix}$$

If fully correlated

$$\sigma_{f_{x,fc}}^2 = [\Delta z_1 \quad \Delta z_2] \begin{bmatrix} \sigma_{\phi_{x1}}^2 & \sigma_{\phi_{x1}} \sigma_{\phi_{x2}} \\ \sigma_{\phi_{x1}} \sigma_{\phi_{x2}} & \sigma_{\phi_{x2}}^2 \end{bmatrix} \begin{bmatrix} \Delta z_1 \\ \Delta z_2 \end{bmatrix},$$

$$\sigma_{f_{x,fc}}^2 = \Delta z_1^2 \sigma_{\phi_{x1}}^2 + 2\Delta z_1 \sigma_{\phi_{x1}} \Delta z_2 \sigma_{\phi_{x2}} + \Delta z_2^2 \sigma_{\phi_{x2}}^2 = (\sigma_{\phi_{x1}} \Delta z_1 + \sigma_{\phi_{x2}} \Delta z_2)^2.$$

$$\sigma_{f_{x,fc}} = \sigma_{\phi_{x1}} \Delta z_1 + \sigma_{\phi_{x2}} \Delta z_2,$$

The standard deviation is a weighted sum of the force densities standard deviations.

If fully uncorrelated

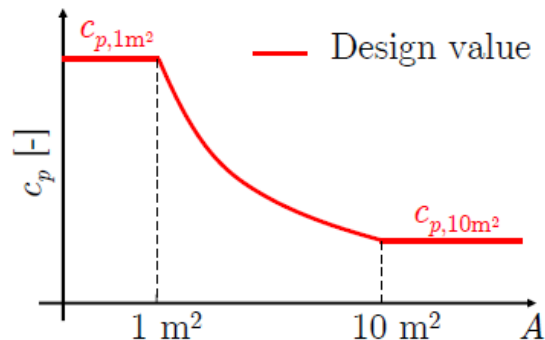
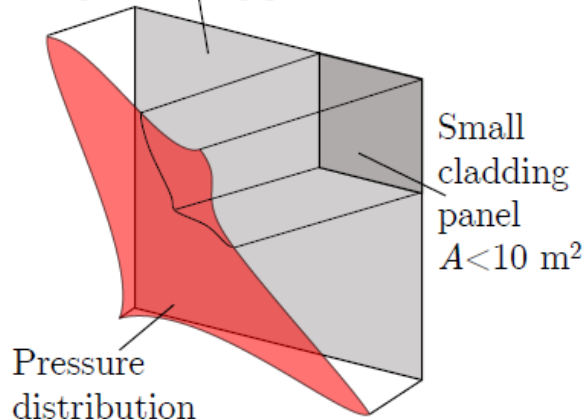
$$\underline{\Sigma}_{\phi_x} = \begin{bmatrix} \sigma_{\phi_{x1}}^2 & 0 \\ 0 & \sigma_{\phi_{x2}}^2 \end{bmatrix}.$$

$$\sigma_{f_x, f_u}^2 = (\sigma_{\phi_{x1}} \Delta z_1)^2 + (\sigma_{\phi_{x2}} \Delta z_2)^2,$$

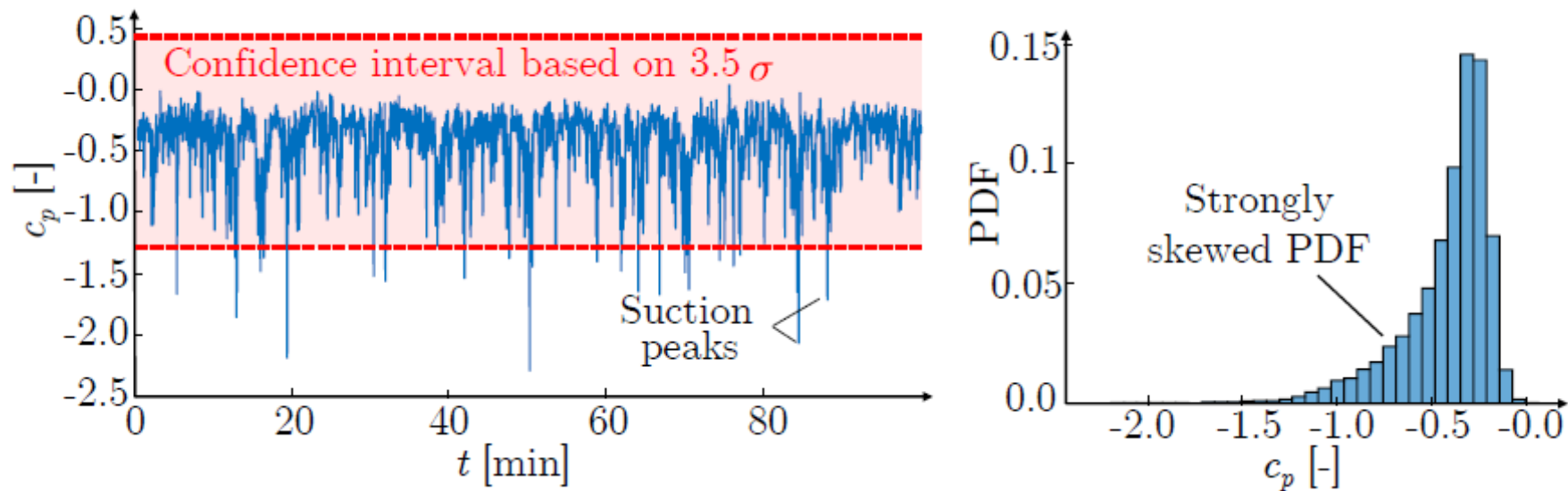
The variance is a weighted sum of the force densities variances.

When there is partial correlation, the result will be intermediate. Such result has deep influence, for instance, in cladding design

Large cladding panel $A > 10 \text{ m}^2$

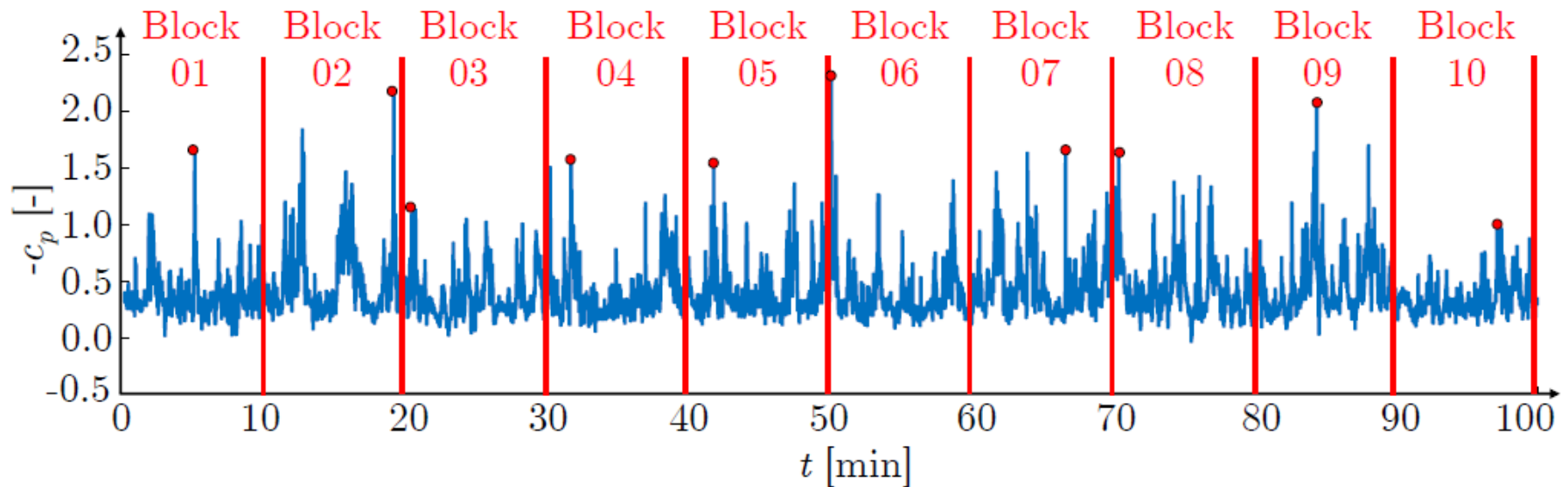


In the previous development, we always assumed Gaussianity. In reality, **numerous responses in wind engineering are non-Gaussian**. A classical example is the cladding pressure.



The problem is quite common every time strong suction are present (e.g. also roof uplift). They show very deep spikes and the PDF is skewed. **The 3.5 sigma confidence interval is often violated!!**

To choose the design value we calculate block maxima. For instance



Based on Extreme Value Analysis, we assume that the block maxima follow a Gumbel distribution.

We can fit the distribution (in the simplest case matching the moments basing on the recorded block-maxima), and use it to evaluate appropriate design values. Usually, the blocks are 10 min long and we extract 80% non exceedance probability values for design.